Boonyakorn Tanrattanakorn

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Education

Chulalongkorn University, BE in Computer Engineering – Bangkok, Thailand Sept 2023 – present

- GPA: 3.93/4.00
- **Relevant Coursework**: Computer Programming (A), Introduction to Data Structures (A), Operating Systems and Programs (A), Database Systems (A), Algorithm Design (A), Hardware Synthesis Laboratory (A), Embedded System Laboratory (A), Software Engineering (A), Introduction to Robotics (A), Computer Networks I (A)

Experience

Embedded System Engineer Intern, Ivory Electronics Co. Ltd. – Thailand June 2026 – present

- Architected a multilayered Arduino firmware (driver → manager → app → UI) for a hydrogen electrolysis teaching instrument, acquiring hydrogen separation current, half-cell voltage, and total dissolved solids via **ADC**.
- Diagnosed a faulty voltage reference IC causing ADC drift; implemented a **median filter** to stabilise readings and restore measurement accuracy.
- Designed a 3D-printed instrument enclosure in **Onshape**, including sensor mounts and wiring layout.

Projects

Project CURSR-III 2023

Chulalongkorn University High Altitude Research Club | Avionics Engineer

- Designed and developed a telemetry ground station using **ESP32** and **LoRa**.
- Designed and assembled 2-layer throughhole **PCB** for ground station using **KiCad**.
- Participated in the fly-off team at the **Spaceport America Cup 2024** in USA.

Project CURSR-IV 2024

Chulalongkorn University High Altitude Research Club | Avionics Engineer

- Designed and assembled a 4-layer SMT flight computer **PCB** using **KiCad**.
- Developed embedded firmware on **STM32 (C/C++)** for sensor integration, telemetry, and flight control.
- Built a **LoRa**-based mesh network ground station using **ESP32** and MQTT for telemetry diversity.
- Modeled and simulated an active airbrake control system in **Simulink**.
- Participated in the fly-off team at the **International Rocket Engineering Competition 2025** in USA.

Project XSRT 2025

Chulalongkorn University High Altitude Research Club | Electrical Engineer

- Developed an R&D web application using **Next.js** and **Tailwind CSS** for a hybrid propulsion test bench user interface.
- Deployed **Flask** backend on **Raspberry Pi 5** to control the test bench.

FPGA Real-Time Video Player 2025

Academic Project | Verilog

- Implemented a real-time FPGA (Basys-3) video player in **Verilog** with VGA output.
- Developed an SPI-based SD card controller to stream 320x240 RGB frames into dual-port RAM.

Smoothed Particle Hydrodynamics Simulator 2025

Personal Project | Godot

- Built a real-time 2D fluid dynamics simulator in **Godot Engine** using the SPH computational method written in **GDScript**.
- Implemented GPU-accelerated physics calculations with **GLSL compute shaders** for density and pressure force computations.

Weather Station 2025

Embedded System Lab I (2110366), Chulalongkorn University | Embedded System Programmer

- Developed **STM32** firmware in **C** to interface with DHT11 temperature/humidity, LDR light, and water level sensors via ADC and a custom one-wire protocol.
- Designed a compact binary UART data packet protocol for STM32–**ESP32-S3** communication with checksum validation.
- Implemented **ESP32-S3** firmware to publish sensor data to a cloud **MQTT** broker (Eclipse Mosquitto) and control a servo motor rain gauge reset via **PWM**.

ThamRoi 2025

Software Engineering Course, Chulalongkorn University | Full-Stack Developer

- Built a full-stack matchmaking web application with a **Next.js (React)** frontend and a **Node.js** REST API in **TypeScript**.
- Designed a **PostgreSQL** schema with **Prisma ORM** wrote seed scripts, migration models, and query controllers.
- Implemented **JWT**-based admin authentication with report moderation, resolve/unresolve flows, and dynamic search filtering.

Aircraft Monitoring & Instrumentation (AMIv2) 2026

Chulalongkorn University High Altitude Research Club | Head of Avionics

- Designed a 2-layer **STM32** avionics **PCB** in **KiCad** integrating IMU, barometer, power monitor, and GPS sensors for the Air Cargo Challenge 2026.
- Developed multi-threaded **FreeRTOS** firmware in **C++** with a flight-phase state machine dynamically adjusting logging rates (1 Hz to 100 Hz).
- Implemented real-time in-flight competition scoring calculation and onboard SD card CSV data logging with **USB Mass Storage** mode.

ACC 2026 Flight Telemetry Dashboard 2026

Personal Project | Full-Stack Developer

- Built a browser-based dashboard for exploring Air Cargo Challenge 2026 standings and flight logs, with **3D flight replay** in **Three.js** and altitude/speed charts in **Chart.js**.
- Processed and validated raw flight telemetry logs into structured datasets using **Python** and **pandas**.

Dual-Channel E-Match Ignition Module 2026

CUHAR / XSRT | Hardware & Embedded Systems Engineer

- Designed a dual-channel pyrotechnic ignition **PCB** in **KiCad** with a capacitive discharge bank and an ignition arming safety.
- Developed embedded firmware on **STM32 (C++)** to communicate as a slave module over a 500 kbit/s **CAN bus**.
- Diagnosed and fixed voltage spikes puncturing low-side MOSFET gate oxide using thermal imaging.

Technical Skills

Embedded Systems: Integrated Circuits, SMT, PCB Design & Assembly, STM32 & ESP32 Programming, FPGA Programming, FreeRTOS, CAN Bus

Programming: Python, C, C++, Java, Verilog

CAD: Fusion 360, KiCad, Onshape

Languages: Thai: Native, English: IELTS 7.5, Japanese: JLPT N4